

Ultra Low Current Consumption

500mA CMOS Voltage Regulator

CE6232 Series

■ INTRODUCTION

The CE6232 series are a group of positive voltage regulators manufactured by CMOS technologies with ultra low power consumption and low dropout voltage, which provide large output currents even when the difference of the input-output voltage is small. The CE6232 series can deliver 500mA output current and allow an input voltage as high as 8V. The series are very suitable for the battery-powered equipments, such as RF applications and other systems requiring a quiet voltage source.

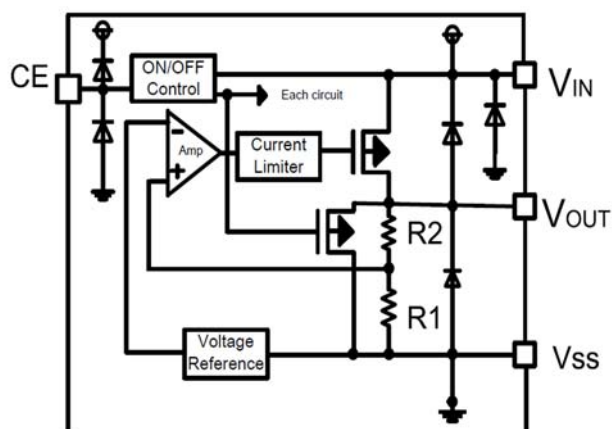
■ FEATURES

- Low Quiescent Current: 0.8 μ A
- Operating Voltage Range: 1.8V~8V
- Output Current: 500mA
- Low Dropout Voltage:
110mV@100mA($V_{OUT}=3.3V$)
- Output Voltage: 1.2~ 5.0V
- High Accuracy: $\pm 2\%/\pm 1\%$ (Typ.)
- High Power Supply Rejection Ratio:
50dB@1kHz
- Low Output Noise:
27x V_{OUT} μ V_{RMS} (10Hz~100kHz)
- Excellent Line and Load Transient Response
- Built-in Current Limiter, Short-Circuit Protection

■ APPLICATIONS

- Portable consumer equipments
- Radio control systems
- Laptop, Palmtops and PDAs
- Wireless Communication Equipments
- Portable Audio Video Equipments
- Ultra Low Power Microcontroller

■ BLOCK DIAGRAM



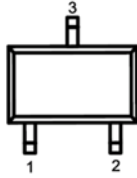
■ ORDER INFORMATION

CE6232①②③④⑤

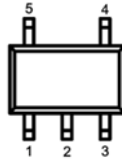
DESIGNATOR	SYMBOL	DESCRIPTION
①	A	Standard
	B	High Active, pull-down resistor built in, with C_{OUT} discharge resistor
②③	Integer	Output Voltage e.g. 1.8V=②:1, ③:8
④	F	Package:DFN1X1-4
	M/ MC/ MY	Package:SOT-23-3
	M	Package:SOT-23-5
	P/PT/PL	Package:SOT-89-3 Package:SOT-89-5
	T/TA/TB	Package:TO-92
⑤	-	2% Accuracy
	1	1% Accuracy

■ PIN CONFIGURATION

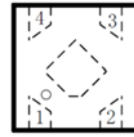
SOT-23-3



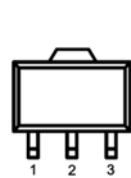
SOT-23-5



DFN1×1-4(Top View)



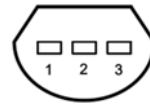
SOT-89-3



SOT-89-5



TO-92



CE6232AXX

PIN NUMBER									PIN NAME	FUNCTION
SOT-23-3			SOT-89-3			TO-92				
M	MC	MY	P	PT	PL	T	TA	TB		
1	3	3	1	2	2	1	2	2	V _{SS}	Ground
2	2	1	3	1	3	3	3	1	V _{OUT}	Output
3	1	2	2	3	1	2	1	3	V _{IN}	Power input

CE6232BXXM (SOT-23-5)

PIN NUMBER	SYMBOL	FUNCTION
1	V _{IN}	Power Input Pin
2	V _{SS}	Ground
3	CE	Chip Enable Pin
4	NC	No Connection
5	V _{OUT}	Output Pin

CE6232BXXF (DFN1×1-4)

PIN NUMBER	SYMBOL	FUNCTION
F		
1	V _{OUT}	Output Pin
2	V _{SS}	Ground
3	CE	Chip Enable Pin
4	V _{IN}	Power Input Pin
EP	Thermal PAD	Ground

CE6232BXXP (SOT-89-5)

PIN NUMBER	SYMBOL	FUNCTION
P		
1	V _{OUT}	Output Pin
2	V _{SS}	Ground
3	NC	No Connection
4	CE	Chip Enable Pin
5	V _{IN}	Power Input Pin

■ ABSOLUTE MAXIMUM RATINGS⁽¹⁾(Unless otherwise specified, T_A=25°C)

PARAMETER		SYMBOL	RATINGS	UNITS
Input Voltage ⁽²⁾		V _{IN}	-0.3~9	V
Output Voltage ⁽²⁾		V _{OUT}	- 0.3~V _{IN} +0.3	V
Output Current		I _{OUT}	600	mA
Power Dissipation	SOT-23	P _D	400	mW
	DFN1×1-4		400	mW
	SOT-89		600	mW
	TO-92		500	mW
Operating Junction Temperature Range		T _j	-40~125	°C
Storage Temperature		T _{stg}	-40~125	°C
Lead Temperature(Soldering, 10 sec)		T _{solder}	260	°C

Note:

(1) Stresses beyond those listed under absolute maximum ratings may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under recommended operating conditions is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

(2) All voltages are with respect to network ground terminal.

■ RECOMMENDED OPERATING CONDITIONS

PARAMETER	MIN.	NOM.	MAX.	UNITS
Supply voltage at V _{IN}	1.8		8	V
Operating junction temperature range, T _j	-40		125	°C
Operating free air temperature range, T _A	-40		85	°C

■ ELECTRICAL CHARACTERISTICS

CE6232 Series

(V_{IN}=V_{OUT}+1V, C_{IN}=C_{OUT}=1μF, T_A=25°C, unless otherwise specified)

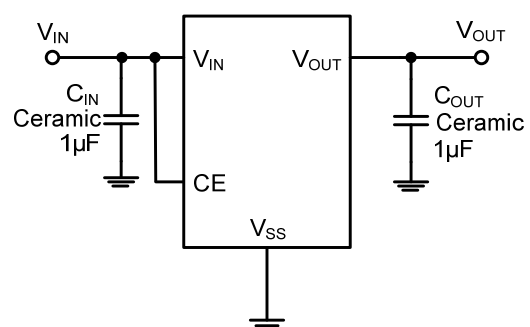
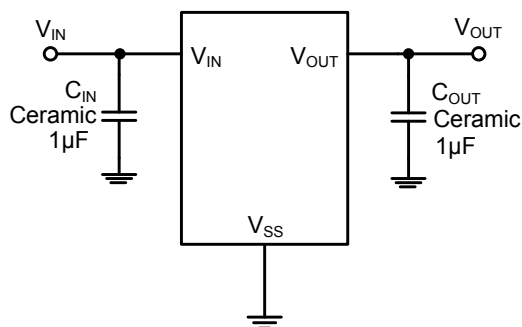
PARAMETER	SYMBOL	CONDITIONS	MIN.	TYP. ⁽³⁾	MAX.	UNITS
Input Voltage	V _{IN}		1.8	—	8	V
Output Voltage Range	V _{OUT}		1.2	—	5	V
DC Output Accuracy		I _{OUT} =1mA	-2	—	2	%
			-1	—	1	%
Dropout Voltage	V _{dif} ⁽⁴⁾	I _{OUT} =100mA, V _{OUT} =3.3V	—	110	—	mV
Supply Current	I _{SS}	I _{OUT} =0, 1.2V≤V _{OUT} ≤3.3V	—	0.8	1.3	μA
		I _{OUT} =0, 3.3V<V _{OUT} ≤5.0V	—	1.0	1.5	μA
Standby Current	I _{STBY}	CE=V _{SS}	—	—	0.2	μA
Line Regulation	$\frac{\Delta V_{OUT}}{V_{OUT} \times \Delta V_{IN}}$	I _{OUT} =10mA V _{OUT} +1V≤V _{IN} ≤8V	—	0.05	0.3	%/V
Load Regulation	ΔV_{OUT}	V _{IN} = V _{OUT} +1V, 1mA≤I _{OUT} ≤100mA	—	10	—	mV
Temperature Coefficient	$\frac{\Delta V_{OUT}}{V_{OUT} \times \Delta T_A}$	I _{OUT} =10mA, - 40°C<T _A <125°C	—	100	—	ppm
Output Current Limit	I _{LIM}	V _{OUT} = 0.5 x V _{OUT(Normal)} , V _{IN} = 5V	550	700	850	mA
Short Current	I _{SHORT}	V _{OUT} =V _{SS}	—	20	—	mA
Power Supply Rejection Ratio	PSRR	I _{OUT} =50mA	100Hz	70	—	dB
			1kHz	50	—	
			10kHz	40	—	
			100kHz	35	—	
Output Noise Voltage	V _{ON}	BW=10Hz to 100kHz	—	27 x V _{OUT}	—	μV _{RMS}
CE "High" Voltage	V _{CE} "H"		1.5	—	V _{IN}	V
CE "Low" Voltage	V _{CE} "L"		—	—	0.3	V
C _{OUT} Auto-Discharge Resistance	R _{DISCHRG}	V _{IN} =5V, V _{OUT} =3.0V, V _{CE} =V _{SS}	—	200	—	Ω

Note:

(3) Typical numbers are at 25°C and represent the most likely norm.

(4) V_{dif}: The Difference Of Output Voltage And Input Voltage When Input Voltage Is Decreased Gradually Till Output Voltage Equals To 98% Of V_{OUT} (E).

■ TYPICAL APPLICATION



■ APPLICATION INFORMATION

Selection of Input/ Output Capacitors

In general, all the capacitors need to be low leakage. Any leakage the capacitors have will reduce efficiency, increase the quiescent current.

A recent trend in the design of portable devices has been to use ceramic capacitors to filter DC-DC converter inputs. Ceramic capacitors are often chosen because of their small size, low equivalent series resistance (ESR) and high RMS current capability. Also, recently, designers have been looking to ceramic capacitors due to shortages of tantalum capacitors. Unfortunately, using ceramic capacitors for input filtering can cause problems. Applying a voltage step to a ceramic capacitor causes a large current surge that stores energy in the inductances of the power leads. A large voltage spike is created when the stored energy is transferred from these inductances into the ceramic capacitor. These voltage spikes can easily be twice the amplitude of the input voltage step.

Many types of capacitors can be used for input bypassing, however, caution must be exercised when using multilayer ceramic capacitors (MLCC). Because of the self-resonant and high Q characteristics of some types of ceramic capacitors, high voltage transients can be generated under some start-up conditions, such as connecting the LDO input to a live power source. Adding a 3Ω resistor in series with an X5R ceramic capacitor will minimize start-up voltage transients.

The LDO also requires an output capacitor for loop stability. Connect a 1μF tantalum capacitor from OUT to GND close to the pins. For improved transient response, this output capacitor may be ceramic.

C_{OUT} Auto-Discharge Function

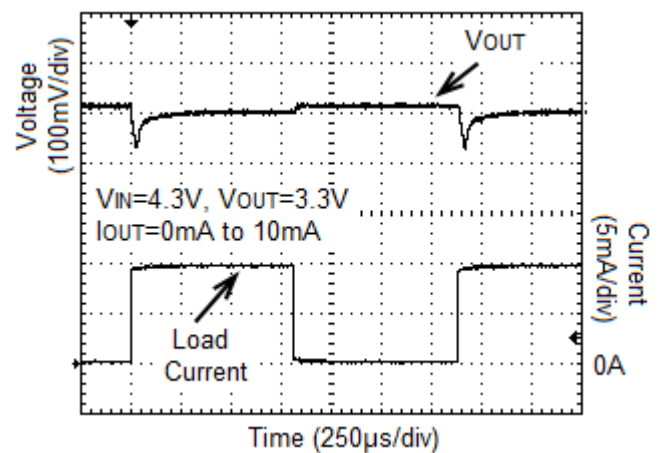
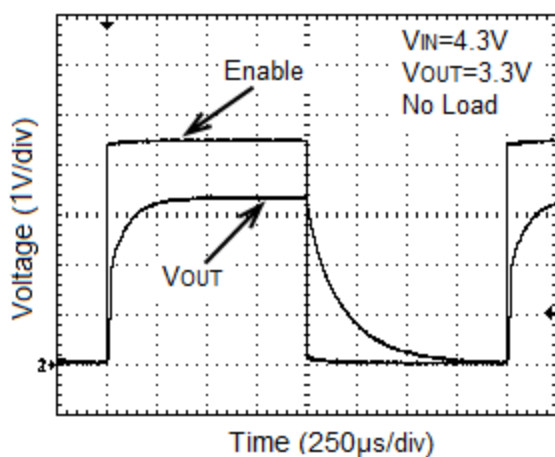
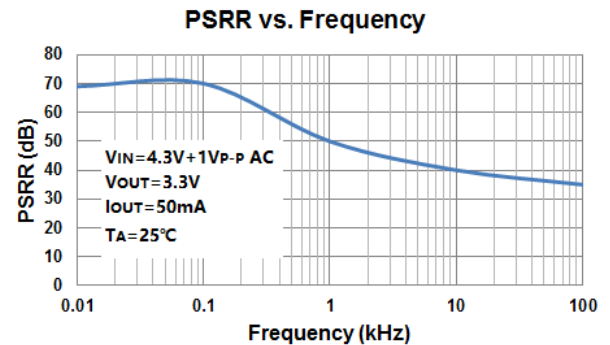
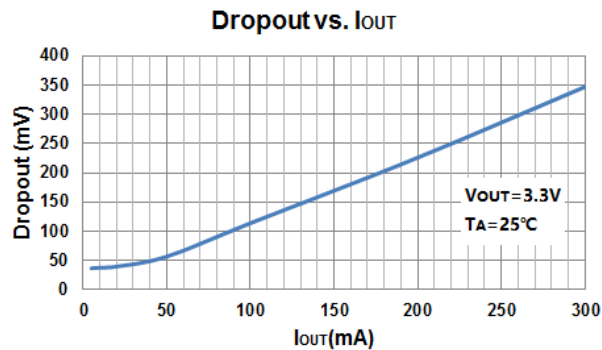
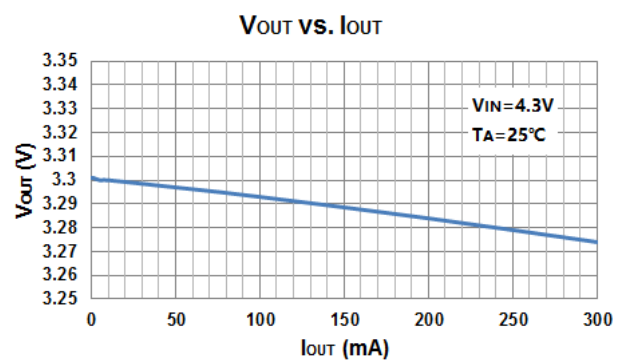
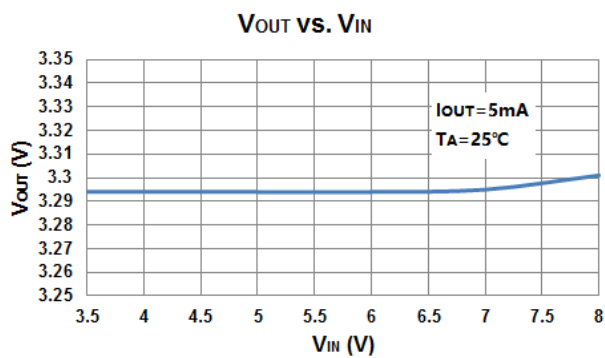
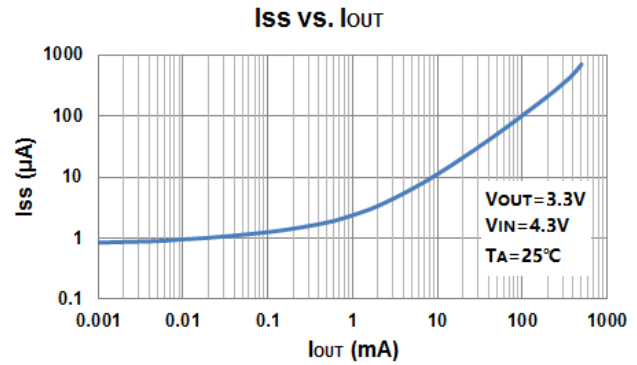
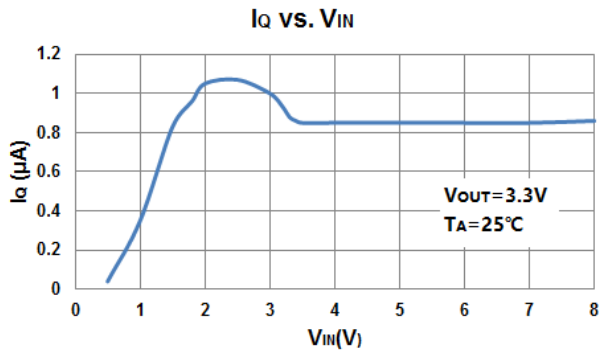
CE6232B series can discharge the electric charge in the output capacitor (C_{OUT}), when a low signal to the CE pin, which enables a whole IC circuit turn off, is inputted via the N-channel transistor located between the V_{OUT} pin and the V_{SS} pin (cf. BLOCK DIAGRAM). The C_{OUT} auto-discharge resistance value is set at 200Ω (V_{OUT}=3.0V @ V_{IN}=5.0V at typical). The discharge time of the output capacitor (C_{OUT}) is set by the C_{OUT} auto-discharge resistance (R) and the output capacitor (C_{OUT}). By setting time constant of a C_{OUT} auto-discharge resistance value [R_{DISCHRG}] and an output capacitor value (C_{OUT}) as τ (τ=C x R_{DISCHRG}), the output voltage after discharge via the N-channel transistor is calculated by the following formulas.

$$V = V_{OUT(E)} \times e^{-t/\tau}, \text{ or } t = \tau/n \ (V/V_{OUT(E)})$$

(V: Output voltage after discharge, V_{OUT (E)}: Output voltage, t: Discharge time,
τ: C_{OUT} auto-discharge resistance R_{DISCHRG}×Output capacitor (C_{OUT}) value C)

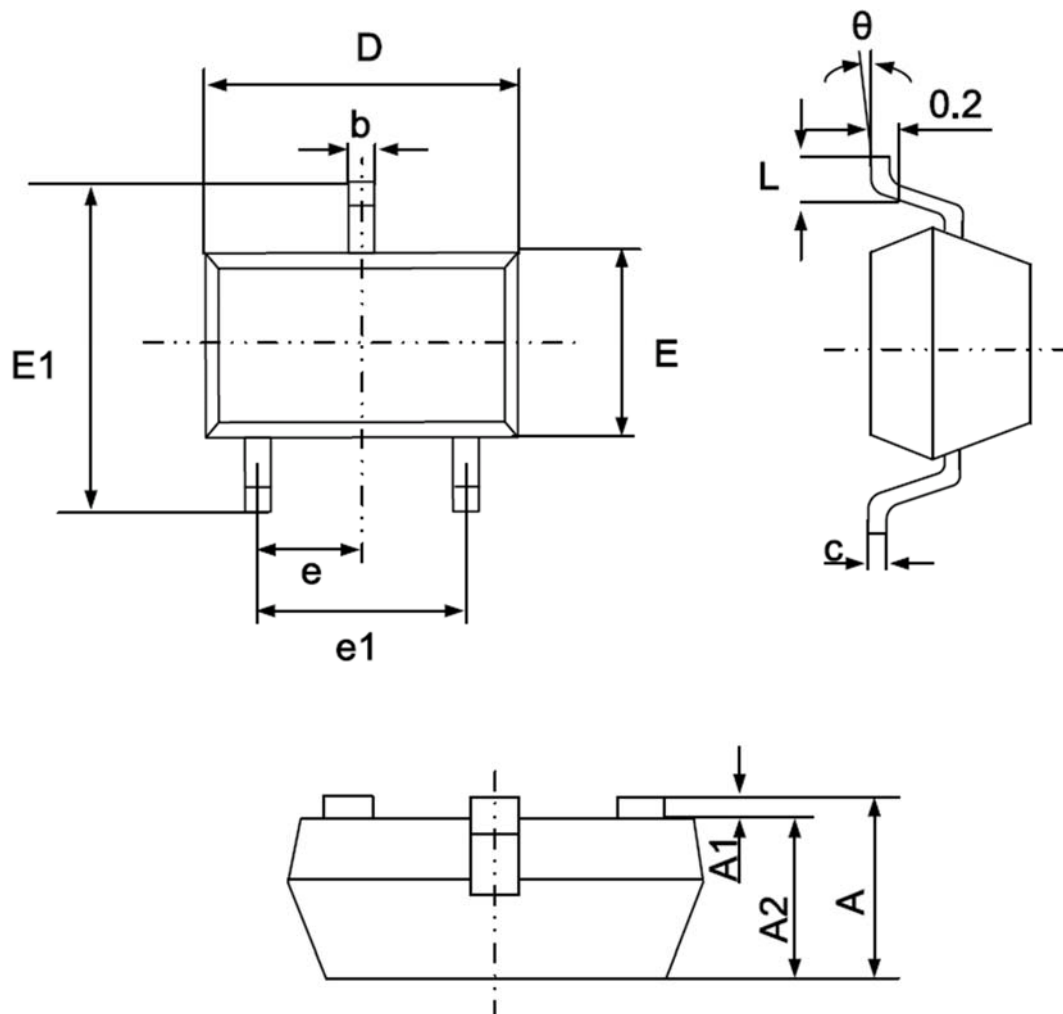
TYPICAL PERFORMANCE CHARACTERISTICS

($V_{IN}=V_{OUT}+1V$, $C_{IN}=C_{OUT}=1\mu F$, $T_A=25^\circ C$, unless otherwise specified)



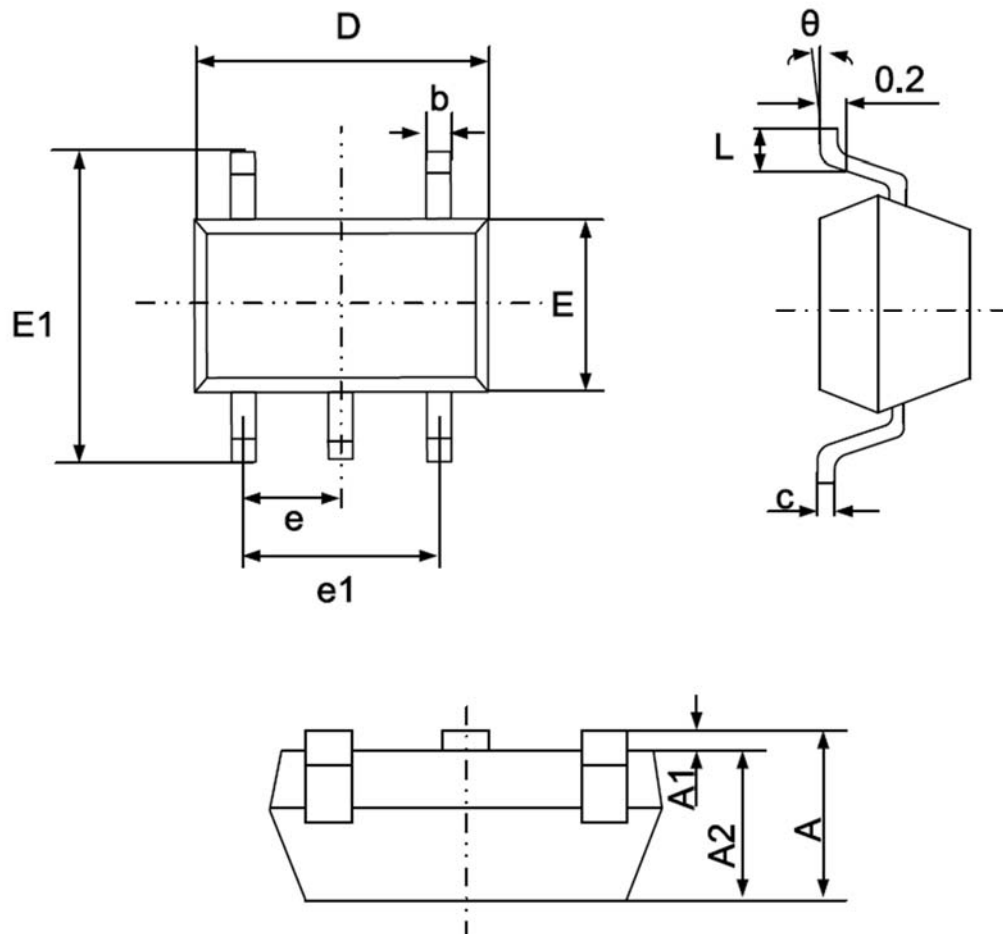
■ PACKAGING INFORMATION

● SOT-23-3 PACKAGE OUTLINE DIMENSIONS



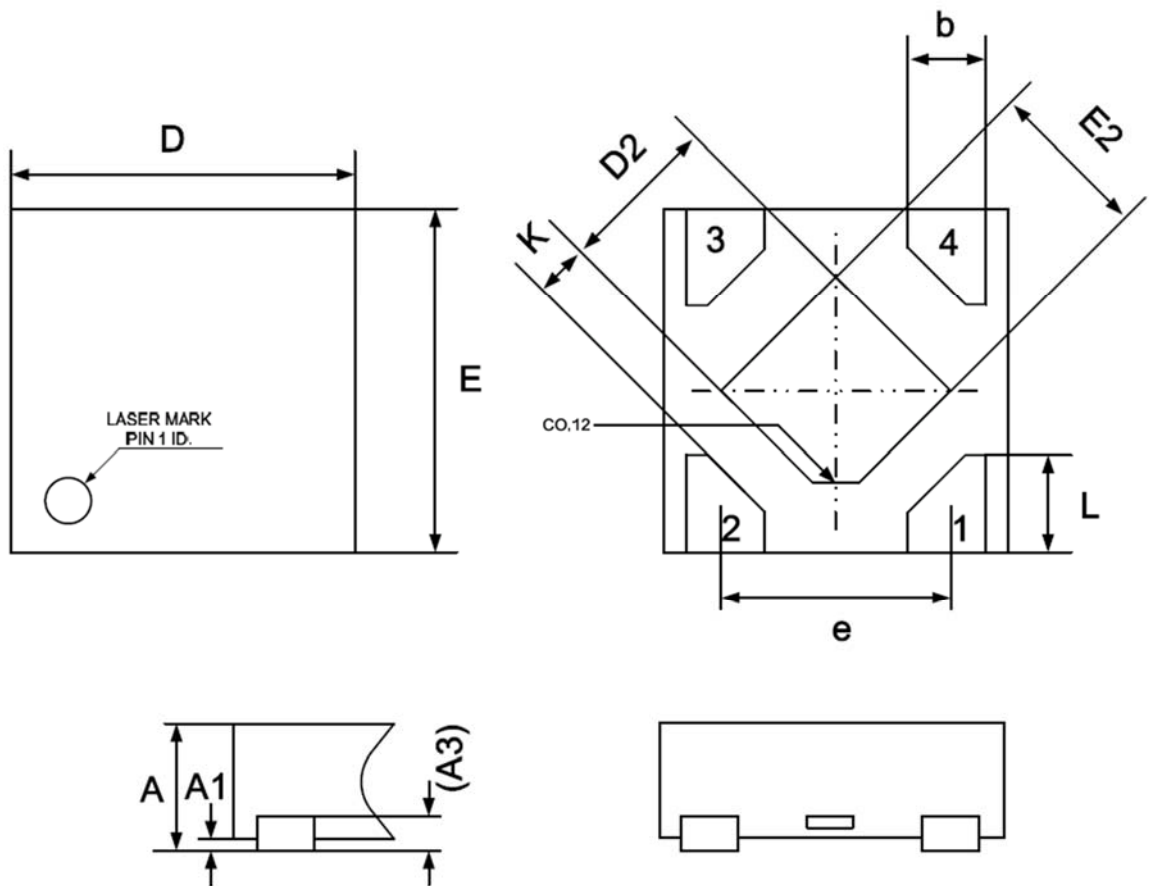
Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min.	Max.	Min.	Max.
A	1.050	1.250	0.041	0.049
A1	0.000	0.100	0.000	0.004
A2	1.050	1.150	0.041	0.045
b	0.300	0.500	0.012	0.020
c	0.100	0.200	0.004	0.008
D	2.820	3.020	0.111	0.119
E	1.500	1.700	0.059	0.067
E1	2.650	2.950	0.104	0.116
e	0.950(BSC)		0.037(BSC)	
e1	1.800	2.000	0.071	0.079
L	0.300	0.600	0.012	0.024
θ	0°	8°	0°	8°

● SOT-23-5 PACKAGE OUTLINE DIMENSIONS



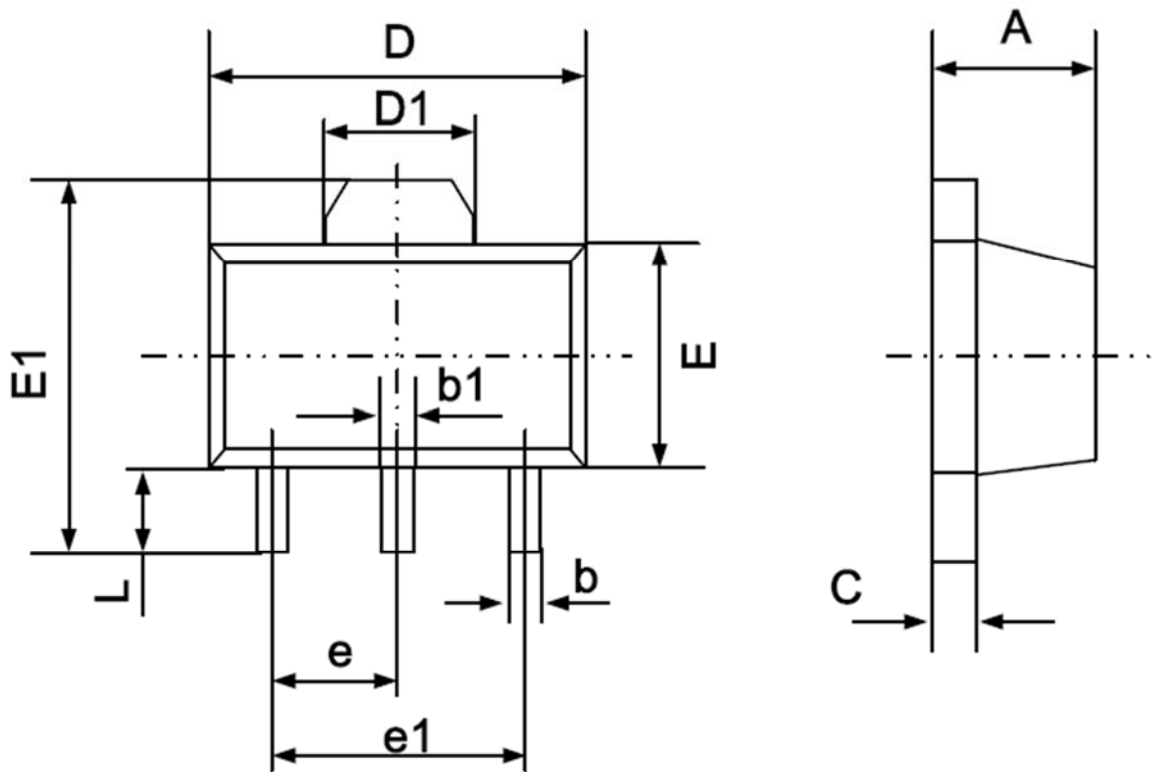
Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min.	Max.	Min.	Max.
A	1.050	1.250	0.041	0.049
A1	0.000	0.100	0.000	0.004
A2	1.050	1.150	0.041	0.045
b	0.300	0.500	0.012	0.020
c	0.100	0.200	0.004	0.008
D	2.820	3.020	0.111	0.119
E	1.500	1.700	0.059	0.067
E1	2.650	2.950	0.104	0.116
e	0.950(BSC)		0.037(BSC)	
e1	1.800	2.000	0.071	0.079
L	0.300	0.600	0.012	0.024
θ	0°	8°	0°	8°

● DFN1X1-4 PACKAGE OUTLINE DIMENSIONS



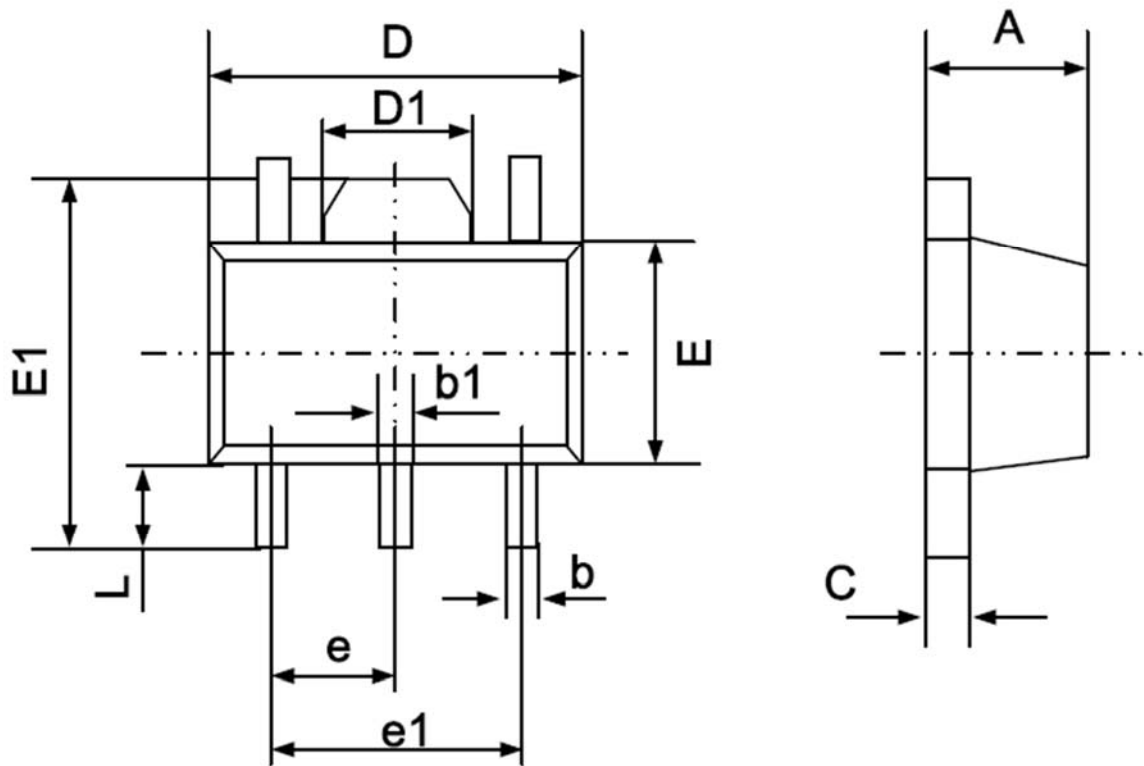
Symbol	Dimensions In Millimeters		
	Min.	Nom.	Max.
A	0.340	0.370	0.400
A1	0.000	0.020	0.050
A3	0.100REF		
b	0.170	0.220	0.270
D	0.950	1.000	1.050
E	0.950	1.000	1.050
D2	0.430	0.480	0.530
E2	0.430	0.480	0.530
L	0.200	0.250	0.300
e	-	0.650	-
K	0.150	-	-

- SOT-89-3 PACKAGE OUTLINE DIMENSIONS



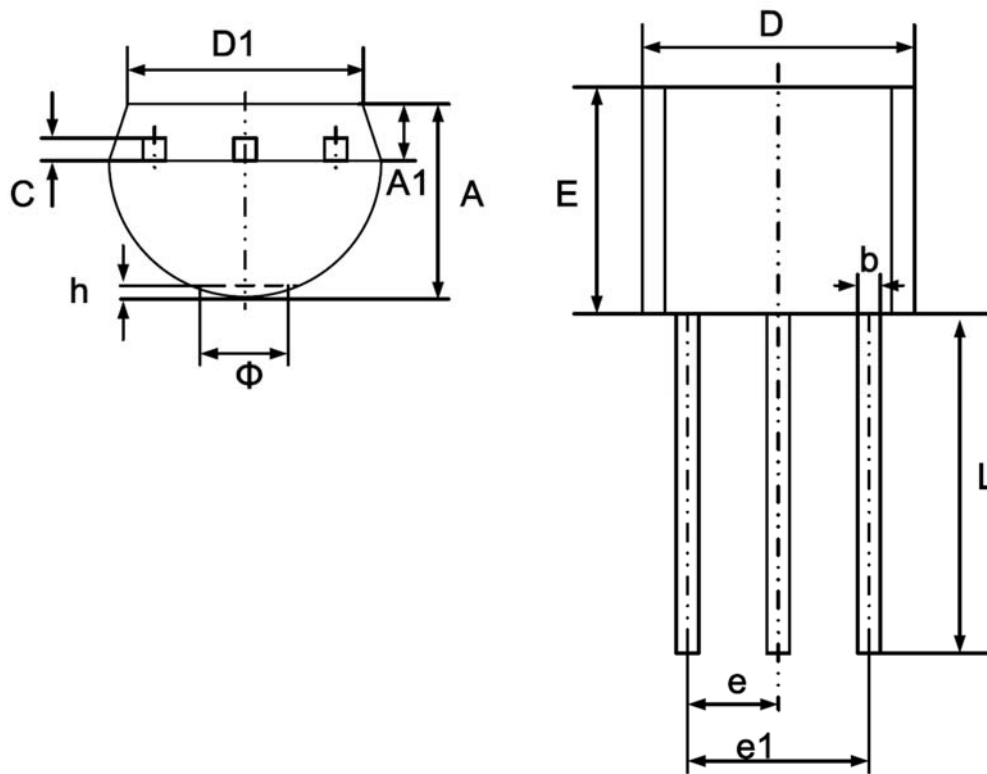
Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min.	Max.	Min.	Max.
A	1.400	1.600	0.055	0.063
b	0.320	0.520	0.013	0.020
b1	0.400	0.580	0.016	0.023
c	0.350	0.440	0.014	0.017
D	4.400	4.600	0.173	0.181
D1	1.550 REF		0.061 REF	
E	2.300	2.600	0.091	0.102
E1	3.940	4.250	0.155	0.167
e	1.500 TYP		0.060 TYP	
e1	3.000 TYP		0.118 TYP	
L	0.900	1.200	0.035	0.047

- SOT89-5 PACKAGE OUTLINE DIMENSIONS



Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min	Max	Min	Max
A	1.400	1.600	0.055	0.063
b	0.320	0.520	0.013	0.020
b1	0.360	0.560	0.014	0.022
c	0.350	0.440	0.014	0.017
D	4.400	4.600	0.173	0.181
D1	1.400	1.800	0.055	0.071
E	2.300	2.600	0.091	0.102
E1	3.940	4.250	0.155	0.167
e	1.500 TYP		0.060 TYP	
e1	2.900	3.100	0.114	0.122
L	0.900	1.100	0.035	0.043

● TO-92 PACKAGE OUTLINE DIMENSIONS



Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min.	Max.	Min.	Max.
A	3.300	3.700	0.130	0.146
A1	1.100	1.400	0.043	0.055
b	0.380	0.550	0.015	0.022
c	0.360	0.510	0.014	0.020
D	4.400	4.700	0.173	0.185
D1	3.430		0.135	
E	4.300	4.700	0.169	0.185
e	1.270 TYP		0.050 TYP	
e1	2.440	2.640	0.096	0.104
L	14.100	14.500	0.555	0.571
Φ		1.600		0.063
h	0.000	0.380	0.000	0.015

© Nanjing Chipower Electronics Inc.

Chipower cannot assume responsibility for use of any circuitry other than circuitry entirely embodied in a Chipower product. No circuit patent license, copyrights or other intellectual property rights are implied. Chipower reserves the right to make changes to their products or specifications without notice. Customers are advised to obtain the latest version of relevant information to verify, before placing orders, that information being relied on is current and complete.